

development cycle, and with software defining most of the product's functionality, embedded software validation is also becoming an important focus area. This is made possible with a new approach called 'in-loop testing.'

From the user perspective, the earlier manual testing methodology had error and cost drawbacks. Since the system was manual, the arrangement always presented gauge repeatability and reproducibility (GRR) errors in the system. In addition, operator costs were directionally proportional to T&M complexities.

"Nowadays, T&M automation solutions have evolved to reasonably low-cost, semi-automatic and fully automatic solutions. Users are seeing this as a boon for their T&M requirements. These tools not only save manpower costs incurred by users but also help in ensuring 100 per cent accuracy with increased throughput," explains T. Anand, managing director, Knewron.

Increased flexibility

General-purpose operating systems like Windows and Linux, coupled with affordable, easily available performance hardware, have increased the flexibility of off-the-shelf instruments, leading to customised and application-oriented solutions.

"Providing test application solutions customised or otherwise is one domain where product companies look to leverage and extend their technical expertise to provide more than just the products to their end users. The term 'solution' is not well defined, since the application requirement of one customer may be different from another. Solution would normally mean taking necessary steps to get the customer to where he intended to be in the first place when he decided

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“**Many of the calibration and verification stages in latest electronic equipment can only be performed through a tight interaction of test equipment, software adjustment and device control, which more or less mandates automated processes**”

— K. Stephen Hire, GM, Aeroflex Asia

Features to look for in T&M automation

While there are certain basic needs that every T&M automation solution should meet, some features are critical from an augmentation and futuristic point of view:

1. Real-time monitoring capabilities
2. Faster response capabilities
3. Programmability
4. Scalability
5. Reliability – repeatability & reproducibility (R&R)
6. Portability

These features not only help design firms in differentiating their products and solutions but also provide larger support in future applications, scale-ups and servicing. They substantially improve automation quality and reduce costs.

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to procure your test instruments. This will, at times, require development of a tailor-made application where engineers from the product company need to work closely with engineers from the end user company," explains Navjodh Dhillon, application engineer, Agilent Technologies.

Mohanram adds, "Most standalone instruments cannot change their functionality as fast as changes in the device under test (DUT) due to the fixed user interface and firmware that must be developed and embedded in the instrument. Thus test engineers are turning to a software-defined approach to instrumentation so that they can quickly customise their equipment to specific application needs and integrate testing directly into the design process. Not only does this approach provide flexibility, it also gives test engineers the ability to take advantage of the performance provided by the latest PC, CPU

and bus technologies."

The new trend of utilising tools based on .NET framework, ActiveX and Java has allowed developers to target all common operating systems like Windows, Linux and more, enabling rapid application development by leveraging the huge amount of existing knowledge already available.

"This has enabled development of well-tested, stable and reliable applications built on proven frameworks. Apart from the available graphical programming environments, languages like C# and .NET provide just the right balance between flexibility and ease of use. Programming with text-based languages is not difficult anymore with most IDEs supporting automatic skeleton code generation, cutting down development time and effort and enabling rapid creation of applications for customisation and test automation. Test developers can take advantage of powerful hardware, including multicore processors in general-purpose computers, to take application performance to unprecedented levels," adds Dhillon.

Standards bring order to chaos

Industry standards have brought a new level of consistency in software and hardware design, and taken un-